Mol Biol Rep (2011) 38:4893–4901
DOI 10.1007/s11033-010-0631-z

Epistatic interactions between loci of one-carbon metabolism modulate susceptibility to breast cancer

Shaik Mohammad Naushad · Addepalli Pavani · Raghunadha Rao Digumarti · Suryanarayana Raju Gottumukkala · Vijay Kumar Kutala

Received: 14 September 2010 / Accepted: 3 December 2010 / Published online: 14 December 2010

Abstract In view of growing body of evidence substantiating the role of aberrations in one-carbon metabolism in the pathophysiology of breast cancer and lack of studies on gene–gene interactions, we investigated the role of dietary micronutrients and eight functional polymorphisms of one-carbon metabolism in modulating the breast cancer risk in 244 case–control pairs of Indian women and explored possible gene–gene interactions using Multifactor dimensionality reduction analysis (MDR). Dietary micronutrient status was assessed using the validated Food Frequency Questionnaire. Genotyping was done for glutamate carboxypeptidase II (GCPII) C1561T, reduced folate carrier (RFC)1 G80A, cytosolic serine hydroxymethyltransferase (cSHMT) C1420T, thymidylate synthase (TYMS) 5′-UTR tandem repeat, TYMS 3′-UTR ins6/del6, methylenetetrahydrofolate reductase (MTHFR) C677T, methyltetrahydrofolate-homocysteine methyltransferase (MTR) A2756G, methyltetrahydrofolate-homocysteine methyltransferase reductase (MTRR) A66G polymorphisms by using the PCR-RFLP/AFLP methods. Low dietary folate intake ($P < 0.001$), RFC1 G80A (OR: 1.38, 95% CI 1.06–1.81) and MTHFR C677T (OR: 1.74 (1.11–2.73) were independently associated with the breast cancer risk whereas cSHMT C1420T conferred protection (OR: 0.72, 95% CI 0.55–0.94). MDR analysis demonstrated a significant tri-variate interaction among RFC1 80, MTHFR 677 and TYMS 5′-UTR loci ($P_{trend} < 0.02$) with high-risk genotype combination showing inflated risk for breast cancer (OR 4.65, 95% CI 1.77–12.24). To conclude, dietary as well as genetic factors were found to influence susceptibility to breast cancer. Further, the current study highlighted the importance of multi-loci analyses over the single-locus analysis towards establishing the epistatic interactions between loci of one-carbon metabolism modulate susceptibility to the breast cancer.

Keywords Breast cancer · One-carbon metabolism · Dietary micronutrients · Epistasis · Multifactor dimensionality reduction

Introduction

Breast cancer is the most common malignancy among Indian women with a steady increase in its incidence [1]. Highly penetrant mutations account for a small proportion of the breast cancer cases [2]. Several studies were conducted globally, to explore etiological factors contributing to majority of the breast cancer cases [3]. The importance of low penetrant genetic polymorphisms as predictive as well as prognostic markers for multi-factorial disorders has become evident after the elucidation of human genome sequence [4]. Genetic polymorphisms and dietary micronutrients of one-carbon metabolism were studied extensively in different cancers as these factors influence DNA synthesis, repair, and methylation. The principal mechanisms of carcinogenesis by these factors are uracil misincorporation in DNA causing DNA damage [5] or aberrant

S. M. Naushad · A. Pavani · V. K. Kutala (✉)
Department of Clinical Pharmacology & Therapeutics, Nizam's Institute of Medical Sciences, Panjagutta, Hyderabad 500082, Andhra Pradesh, India
e-mail: vijaykutala@gmail.com

R. R. Digumarti
Department of Medical Oncology, Nizam's Institute of Medical Sciences, Panjagutta, Hyderabad 500082, India

S. R. Gottumukkala
Department of Surgical Oncology, Nizam's Institute of Medical Sciences, Panjagutta, Hyderabad 500082, India

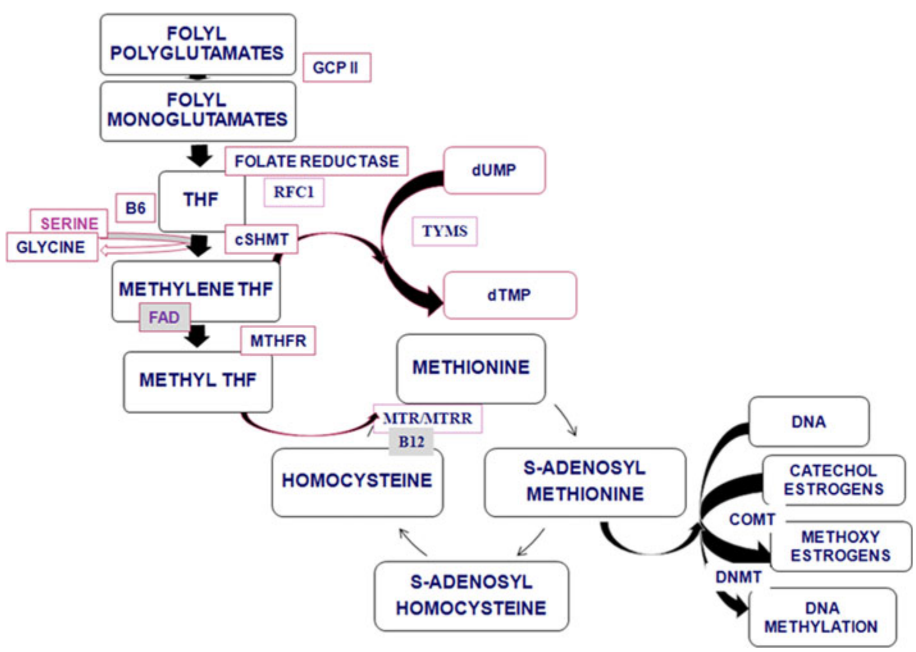
methylation (focal hypermethylation and global hypomethylation) that triggers the inactivation of tumor suppressor genes and inactivation of proto-oncogenes [6].

One-carbon metabolism harbors several crucial biological reactions from folate uptake to synthesis of S-adenosylmethionine (SAM). Glutamate carboxypeptidase II (GCPII, MIM: 600934), reduced folate carrier 1 (RFC1: MIM: 600424), cytosolic serine hydroxymethyltransferase (cSHMT: MIM: 182144), and thymidylate synthase (TYMS, MIM: 188350) enzymes regulate folate uptake, transport across RBC membrane, synthesis of 5,10-methylene tetrahydrofolate, and synthesis of thymidylate, respectively. Methylenetetrahydrofolate reductase (MTHFR, MIM: 607093), 5-methyltetrahydrofolate-homocysteine methyltransferase (MTR, MIM: 156570), and 5-methyltetrahydrofolate-homocysteine methyltransferase reductase (MTRR: MIM: 602568) are the enzymes that regulate the FAD-dependent reduction of 5,10-methylene tetrahydrofolate to 5-methyltetrahydrofolate, synthesis of methionine, and the reductive methylation of cobalamin, respectively. (Scheme 1) Any perturbation in this metabolic pathway can impair the synthesis of SAM, an universal methyl group donor, thereby leading to epigenetic changes, specifically aberrant DNA methylation.

Epidemiological studies have shown an inverse association between breast cancer risk and dietary intake of green vegetables, white vegetables, mushrooms and folate [7]. Till to date, no studies have been reported on the GCPII C1561T (rs61886492) polymorphism, a crucial polymorphism found to influence the intestinal absorption of folate, in relation to the breast cancer. A limited number of studies have been reported on the RFC1 G80A (rs1051266) and cSHMT C1420T (rs1979277) polymorphisms with a relevance to the breast cancer [8, 9]. An association of RFC1 G80A polymorphism with the breast cancer has not been documented [8]. Two studies on the association of cSHMT C1420T polymorphisms offer contradictory findings which warrant further corroboration [8, 9]. TYMS 5′-UTR 28 bp tandem repeat polymorphism showed no association with the breast cancer [8] where as 3′-UTR ins6/del6 showed an inverse association [10].

Conflicting results were obtained regarding the association of MTHFR C677T (rs1801133) polymorphism with the breast cancer risk [11–13]. Low folate intake [7] or carrier status for BRCA1 [14] or prolonged exposure to the estrogens prior to first full term pregnancy [15] has been found to increase the risk for the breast cancer synergistically in the subjects with MTHFR 677 T-allele. MTR A2756G (rs1805087) polymorphism has been reported to reduce the risk for the breast cancer in one study [16] whereas, in another study, no such association has been observed [17]. MTRR A66G (rs1801394) polymorphism has been shown to alter the susceptibility to colorectal cancer [18], acute leukemia [19], lung cancer [20] and squamous cell carcinoma [21], however, no such association has been observed with the breast cancer [17].

All these polymorphisms have been reported to be putatively functional. Two of these polymorphisms, TYMS 5′-UTR 28 bp tandem repeat and TYMS 3′-UTR ins6/del6 are synonymous polymorphisms that influence the transcription of TYMS and stability of the TYMS mRNA, respectively. All other polymorphisms are non-synonymous single nucleotide polymorphisms (SNPs) affecting the enzyme activities [22]. Despite the modern advances in



Scheme 1 One-carbon metabolism. This scheme illustrates the vital biological reactions starting from folate uptake to synthesis of S-adenosylmethionine. *GCPII* glutamate carboxypeptidase II; *RFC1* reduced folate carrier 1; *cSHMT* cytosolic serine hydroxymethyltransferase; *TYMS* thymidylate synthase; *MTHFR* methylenetetrahydrofolate reductase; *MTR* methyltetrahydrofolate-homocysteine methyltransferase; *MTRR* methyltetrahydrofolate-homocysteine methyltransferase reductase; *dUMP* Uracil monophosphate; *dTMP* thymine monophosphate

human genetics, no comprehensive multi-locus studies on Indian women in relation to the breast cancer have been documented. Results on the single locus investigations are also sparse [23]. Furthermore, wide variations at the ethnic and population levels have been reported in the distribution of these polymorphisms [22]. In view of this, we conducted a case–control study on South Indian subjects to examine the individual and combined effects of eight genetic polymorphisms such as GCPII C1561T, RFC1 G80A, cSHMT C1420T, TYMS 5′-UTR tandem repeat, TYMS 3′-UTR ins6/del6, MTHFR C677T, MTR A2756G and MTRR A66G, and the dietary status of four B vitamins in modulating susceptibility to the breast cancer.

Materials and methods

Study population

Eligible cases were a consecutive series of female patients aged 18–80 years with newly diagnosed and histologically confirmed breast cancer. Cases were recruited during the period of January, 2009 to July, 2010 at the Nizam's Institute of Medical Sciences (NIMS), Hyderabad, India. A total of 244 cases met these criteria. Eligible controls were subjects without any history of cancer, selected from a group of 270 healthy women volunteers. These volunteers comprised mostly of hospital staff or their relatives who had no history of benign or malignant breast disease. Their recruitment was done at NIMS, Hyderabad after careful evaluation of their personal health and family history. The selection of controls was done by matching each control to each case by age (±5years), ethnicity, region, linguistic origin and menopausal status. Ultimately, 244 matched case–control pairs were enrolled in the study. Informed consent was obtained from all the subjects. This study was approved by the Institutional Ethical committee of Nizam's Institute of Medical Sciences, Hyderabad (EC/NIMS/767/2007, dated 05.09.2008).

Demographic data collection and dietary assessment

Personal interviews were conducted by trained interviewers using a standardized questionnaire which included demographic characteristics, life style (exercise, smoking, alcohol intake, tobacco chewing), reproductive history (age of menarche, age at the time of first full term pregnancy, breast feeding history, number of live births, number of miscarriages, menopausal status, use of oral contraceptives, hormone replacement therapy) and family history (breast cancer or any other malignancies). Measurements of height and weight were taken at the time of recruitment from all the subjects to calculate body mass index (BMI). Clinical details were obtained from medical records of the patients. Personal interviews were conducted during their first visit to the oncologist after confirmed diagnosis (for cases). The blood samples were collected on the same day following the interview in order to minimize the impact of surgical or therapeutic intervention on biochemical parameters.

For food frequency questionnaire, all the subjects were requested to keep a log on the type of food item consumed, quantity of food consumed, and frequency of food consumption (times per day/week/month/three months or never) over a period of 1 week. Daily nutrient intakes were calculated as grams of food multiplied by the amount of each nutrient in the food and the frequency of consumption, summing over all the foods consumed. The composition of raw and cooked food items was determined from the 2007 report on the *Nutritive value of Indian foods* [24]. For further information on the composition of other food items, *McCance and Widdowson's The composition of Foods* [25] and the United States Department of Agriculture's National Nutrient Database for Standard Reference release 18 (USDA, Washington, DC, USA) were consulted [26]. Subjects who are on vitamin supplementation were not included as no comprehensive data were available.

The validity of micronutrient intake estimated from the FFQ was established by two approaches. A subgroup of 62 women, who visited the hospital for a follow-up, completed one-week dietary record twice. The dietary micronutrient status at two different time intervals was correlated with one another. Spearman rank correlation coefficients for folate, B2, B6 and B12 were 0.35, 0.30, 0.34 and 0.32, respectively. We validated this data further by the biochemical determination of folate in the plasma of subjects who are not on anti-folate medication/folate supplementation. Spearman rank correlation coefficient was 0.29 for dietary folate versus plasma folate levels.

Genetic analysis

Whole blood samples were collected in EDTA from all the subjects and plasma was separated immediately and stored at −80°C for further biochemical analyses. The buffycoat was used for the genomic DNA isolation by the method described by Salazar et al. [27]. The extracted DNA was analyzed for eight genetic polymorphisms using the polymerase chain reaction-restriction fragment length polymorphism (PCR-RFLP) and PCR-amplified fragment length polymorphism (PCR-AFLP) methods. Cases and matched controls were analyzed in the same set of PCR by the analyst blinded to case–control status. For all the genetic analyses, each PCR set was accompanied by a negative control without genomic DNA in order to check contamination of components. Activities of all the restriction enzymes were checked by digesting the plasmids with known restriction sites. For RFLP analysis, a positive control was included in each set, which ensures complete digestion. Genotypes in few specimens were randomly rechecked to rule out the genotyping errors and 100% concordance was observed.

Plasma folate estimation

Plasma folate was determined on the AxSYM automated analyzer (Abbott Laboratories) according to the manufacturer's instructions using the AxSYM folate assay kits. Plasma folate analysis was performed only in the subjects who were not on any anti-folate medication or vitamin supplementation at the time of specimen collection. The intra assay CV was 3.8% where as inter assay CV was 4.2%.

Statistical analysis

Demographic characteristics distributed as categorical variables were subjected to simple logistic regression to assess the trend associated with each characteristic. For other demographic data where only absence or presence of the variable was assessed, Fisher's exact test was employed. All the genetic analysis data were computed in 0, 1 and 2 model representing homozygous wild, heterozygous and homozygous mutant genotypes, respectively. Since the representative samples for homozygous mutant genotype are less for some polymorphisms, the comparisons were made based 0 and 1 + 2 combinations. Conditional logistic regression analysis was done on the control data for confounding effects and adjusted odds ratios (ORs) were obtained. Potential confounders considered for the multivariate analysis were body mass index, age at menarche, age at first full term pregnancy, parity, and the family history of breast cancer. They were treated as continuous variables for logistic regression analysis. Wherever the information was in the format of yes/no, it was labeled as 1/0 respectively. The Hardy–Weinberg equilibrium was checked for cases and controls by adopting the chi-squared test.

All the continuous variables were subjected to the student's *t*-test to obtain mean, standard deviation (SD) and the *P* value. In order to validate the food frequency questionnaire, the plasma folate levels were correlated with the dietary folate levels by using the Spearman rank correlation coefficient. Multifactor dimensionality reduction analysis (MDR, version 2.0 beta 6) was used to reduce the data dimension and to get a set of single nucleotide polymorphisms (SNPs) that can interpret the results best. The data obtained was again subjected to the Fisher's exact test for each genotype combination to ascertain the trend across different combinations. Dietary micronutrient status was stratified into three tertiles. Each variant allele frequency in cases and controls was computed across three tertiles and a trend test was performed to assess gene-nutrient interactions. For all the above statistical analysis, the statistical web page "www.statpages.org" was used. For all associations, a P value of $<0.05/n$ (n = number of variables tested) was considered as statistically significant. All the observed statistically significant genetic associations exhibited ≥80% power with type I-alpha error of 0.05.

Results

Demographic characteristics of cases and controls at baseline are shown in Table 1. Women with low body mass index (BMI) were found to have increased risk. However, a careful evaluation revealed that risk is due to low plasma folate, not directly due to low BMI. However, a significant reduction in breast cancer risk was observed in women who had optimal BMI. No statistically significant association was observed for regular exercise, smoking, alcohol intake, tobacco chewing, age at menarche, parity, age at first full term pregnancy and breast feeding. Family history of breast cancer and low dietary folate intake were associated with the breast cancer risk (Table 1).

Individual genetic effects

The genotype distribution for all the polymorphisms (>0.29) except for MTR A2756G and MTRR A66G was in accordance with the Hardy–Weinberg equilibrium in cases and controls. Among the eight putatively functional polymorphisms tested, RFC1 G80A (OR: 1.38, 95% CI 1.06–1.81) and MTHFR C677T (OR: 1.74 (1.11–2.73) were independently associated with the breast cancer risk, whereas cSHMT C1420T conferred protection (OR: 0.72, 95% CI 0.55–0.94). Other polymorphisms showed no independent effects (Table 2).

Gene–gene interactions

The multifactor dimensionality reduction as well as the logistic regression indicated significant epistatic interactions between three allelic variants i.e. RFC1 80A-MTHFR 677T-TYMS 2R in modulating the breast cancer risk in dose-dependent manner ($P_{trend} < 0.02$) (Table 3). When the different genotype combinations at RFC1, TYMS and MTHFR loci were examined, with reference to the triple wild genotype, presence of TYMS 2R allele alone showed 2.17-fold risk (95% CI 1.06–4.43) for the breast cancer. Co-segregation of RFC1 80A and MTHFR 677T-variant alleles was associated with 2.89-fold risk (95% CI 1.13–7.42) for the breast cancer. When all the three loci

Table 1 Demographic characteristics of breast cancer cases and controls

Variable	Cases (%)	Controls (%)	P
Age in years			
20–29	14 (5.7)	15 (6.1)	
30–39	29 (11.9)	28 (11.5)	
40–49	62 (25.4)	64 (26.2)	
50–59	61 (25.0)	60 (24.6)	
60–69	44 (18.0)	42 (17.2)	
70–79	32 (13.1)	33 (13.5)	
80–89	2 (0.8)	2 (0.8)	0.92
Body mass index (BMI) in kg/m^2			
<18.5	21 (8.6)	8 (3.3)	0.02*
18.5–24.9	85 (34.8)	115 (47.1)	0.008*
≥25.0	138 (56.6)	121 (49.6)	0.15
Regular exercise	40 (16.4)	56 (23.0)	0.07
Smoking	45 (18.4)	51 (20.9)	0.49
Alcohol intake	12 (4.9)	5 (2.0)	0.08
Tobacco chewing	7 (2.9)	3 (1.2)	0.20
Age at menarche in years			
<12	16 (6.6)	38 (15.5)	
12–14	194 (79.5)	160 (65.6)	
>14–16	31(12.7)	45 (18.4)	
>16	3 (1.2)	1 (0.4)	0.28
Parity			
0	26 (10.7)	25 (10.2)	
1–2	106 (43.4)	110 (45.1)	
≥3	112 (45.9)	109 (44.7)	0.89
Age at first full term pregnancy in years			
<18	87 (35.7)	77 (31.6)	
19–20	46 (18.9)	47 (19.3)	
>20–25	54 (22.1)	52 (21.3)	
>25–30	30 (12.3)	40 (16.4)	
>30–35	1 (0.4)	3 (1.2)	0.16
Breast feeding	225 (92.2)	228 (93.4)	0.60
Menopause status			
Pre-menopausal	82 (33.6)	81 (33.2)	
Post-menopausal	162 (66.4)	163 (66.8)	0.92
Family history of breast cancer			
In first-degree relatives	14 (5.7)	4 (1.6)	0.01*
In second-degree relatives	7 (2.9)	3 (1.2)	0.20
Dietary intake of micronutrients			
Folate, µg/day	340 ± 112	377 ± 141	0.001*
Vitamin B2, mg/day	1.27 ± 0.43	1.21 ± 0.47	0.14
Vitamin B6, mg/day	1.51 ± 0.37	1.46 ± 0.40	0.15
Vitamin B12, µg/day	5.99 ± 1.55	5.94 ± 1.26	0.70

For all these univariate analyses *P* value < 0.05 was considered statistically significant. * *P* value is statistically significant

P values were derived based on the trend across different categories. *P* values for one category were based on Fisher's exact test (for categorical variable) and student's *t*-test (for continuous variable)

harbored variant alleles, the risk was found to be much higher, i.e. 4.65-fold (95% CI 1.77–12.24). No gene–gene interactions were observed between cSHMT C1420T and other polymorphisms.

Gene-nutrient interaction

Dietary folate showed no interaction with GCP II ($P_{trend} = 0.37$), RFC ($P_{trend} = 0.14$), cSHMT ($P_{trend} = 0.46$), TYMS 5′-UTR ($P_{trend} = 0.32$), TYMS 3′-UTR ($P_{trend} = 0.30$), MTHFR ($P_{trend} = 0.70$), MTR ($P_{trend} = 0.38$) and MTRR ($P_{trend} = 0.68$) polymorphisms.

Correlation between genotypes and plasma folate levels

Result of the study as shown in Fig. 1, illustrates the functional relevance of GCPII C1561T, RFC1 G80A and cSHMT C1420T polymorphisms. GCPII C1561T (CC vs. CT: 7.25 ± 1.43 ng/ml vs. 6.12 ± 1.98 ng/ml, $P < 0.0001$) and RFC1 G80A (GG: 7.12 ± 1.36 ng/ml, GA: 6.84 ± 1.15 ng/ml and AA: 6.84 ± 1.11 ng/ml, $P < 0.05$) polymorphisms showed inverse association with folate where as cSHMT polymorphism showed positive association (CC:6.23 ± 1.13 ng/ml, CT: 6.86 ± 1.159 ng/ml and TT:7.64 ± 1.08 ng/ml, $P < 0.0001$).

Discussion

The current study investigated the association between the aberrations in one-carbon metabolism and the breast cancer risk in Indian women. This is the first comprehensive study from India in relation to the breast cancer, and we are the first group to investigate the role of GCPII C1561T polymorphism. In this study, we found RFC1 G80A and MTHFR C677T polymorphisms as independent risk factors for the breast cancer. An inverse association was observed between the cSHMT C1420T polymorphism and the breast cancer. None of the other polymorphisms showed any association with the breast cancer. Among the dietary B vitamins, only dietary folate showed an inverse association. Using MDR analysis, trivariate interactions between RFC1/TYMS/MTHFR loci were found to inflate the risk for the breast cancer. Although cSHMT C1420T polymorphism was protective against breast cancer, no specific interaction between any other variant alleles was observed. No significant gene-nutrient interactions were observed. Polymorphisms in GCPII, RFC1 and cSHMT were found to influence the plasma folate pool. GCP II C1561T and RFC1 G80A polymorphisms showed inverse association with the plasma folate where as cSHMT C1420T polymorphism showed positive association with the plasma folate.

Table 2 Distribution of genetic polymorphisms in one-carbon metabolism among the breast cancer cases and controls

Polymorphism	WW (%)	WM (%)	MM (%)	MAF (%)	Adjusted OR (95% CI)
GCPII C1561T	CC	CT	TT	T-allele	
Cases	208 (85.2)	36 (14.8)	0 (0.0)	36/488 (7.4)	0.74 (0.46–1.22)
Controls	195 (80.6)	47 (19.4)	0 (0.0)	47/484 (9.7)	
RFC1 G80A	GG	GA	AA	A-allele	
Cases	87 (35.7)	107 (43.9)	50 (20.5)	207/488 (42.4)	1.38 (1.06–1.81)*
Controls	96 (39.3)	122 (50.0)	26 (10.7)	174/488 (35.7)	
cSHMT C1420T	CC	CT	TT	T-allele	
Cases	56 (23.0)	128 (52.5)	60 (24.6)	248/488 (50.8)	0.72 (0.55–0.94)*
Controls	43 (17.6)	119 (48.8)	82 (33.6)	283/488 (58.0)	
TYMS 5′-UTR 3R/2R	3R3R	2R3R	2R2R	2R-allele	
Cases	99 (40.6)	110 (45.1)	35 (14.3)	180/488 (36.9)	1.08 (0.83–1.42)
Controls	101 (41.4)	117 (48.0)	26 (10.7)	169/488 (34.6)	
TYMS 3′-UTR Ins6/del6	Ins6/Ins6	Del6/Ins6	Del6/Del6	Del6-allele	
Cases	55 (22.5)	122 (50.0)	67 (27.5)	256/488 (52.5)	1.06 (0.81–1.40)
Controls	58 (23.8)	130 (53.3)	56 (23.0)	242/488 (49.6)	
MTHFR C677T	CC	CT	TT	T-allele	
Cases	185 (75.8)	56 (23.0)	3 (1.2)	62/488 (12.7)	1.74 (1.11–2.73)*
Controls	205 (84.0)	39 (16.0)	0 (0.0)	39/488 (8.0)	
MTR A2756G	AA	AG	GG	G-allele	
Cases	117 (48.0)	115 (47.1)	12 (4.9)	139/488 (28.5)	1.06 (0.77–1.45)
Controls	122 (50.0)	112 (45.9)	10 (4.1)	132/488 (27.0)	
MTR A2756G	GG	AG	AA	A-allele	
Cases	61(25.0)	172 (70.5)	11 (4.5)	194/488 (39.8)	1.32 (0.93–1.88)
Controls	78 (32.0)	154 (63.1)	12 (4.9)	178/488 (36.5)	

WW wild genotype; *WM* heterozygous genotype; *MM* homozygous mutant genotype; *MAF* minor allele frequency; *Adjusted OR* Odds ratio adjusted for age, body mass index, age at menarche, age at first full term pregnancy, parity, family history of breast cancer; *CI* confidence interval. * *P* value statistically significant (<0.05)

The association of each genetic variant with breast cancer was elucidated by conditional logistic regression analysis. The potential effect of confounding variables was controlled at each level

Table 3 Epistatic interactions between three loci of one-carbon metabolism modulating susceptibility to breast cancer

RFC-MTHFR-TYMS	Cases (%)	Case/control (%)	Adjusted OR (95% CI)
W-W-W	25 (10.2)	40 (16.4)	1.00 (ref)
W-W-M	45 (18.4)	36 (14.8)	2.17 (1.06–4.43)*
M-W-W	46 (18.9)	44 (18.0)	1.86 (0.92–3.75)
M-W-M	69 (28.3)	85 (34.8)	1.29 (0.70–2.40)
W-M-W	9 (3.7)	7 (2.9)	2.17 (0.66–7.11)
W-M-M	8 (3.3)	13 (5.3)	1.00 (0.35–2.85)
M-M-W	19 (7.8)	10 (4.1)	2.89 (1.13–7.42)*
M-M-M	23 (9.4)	9 (3.7)	4.65 (1.77–12.24)*

W wild genotype; *M* heterozygous and homozygous mutant genotypes; *adjusted OR* Odds ratio adjusted for age, body mass index, age of menarche, parity and family history of breast cancer; *CI* confidence interval; *ref* reference genotype combination. * *P* value statistically significant

The data obtained by multifactor dimensionality reduction analysis was reconfirmed using multiple logistic regression to demonstrate the risk pattern across different genotype combinations

The age distribution as depicted in Table 1, showed doubling of the breast cancer risk in women with each decade of life after the age of 20 years till the age of 50 years, suggesting the role of reproductive hormones in the etiology of breast cancer [28]. Furthermore, post-menopausal women had higher risk for breast cancer,

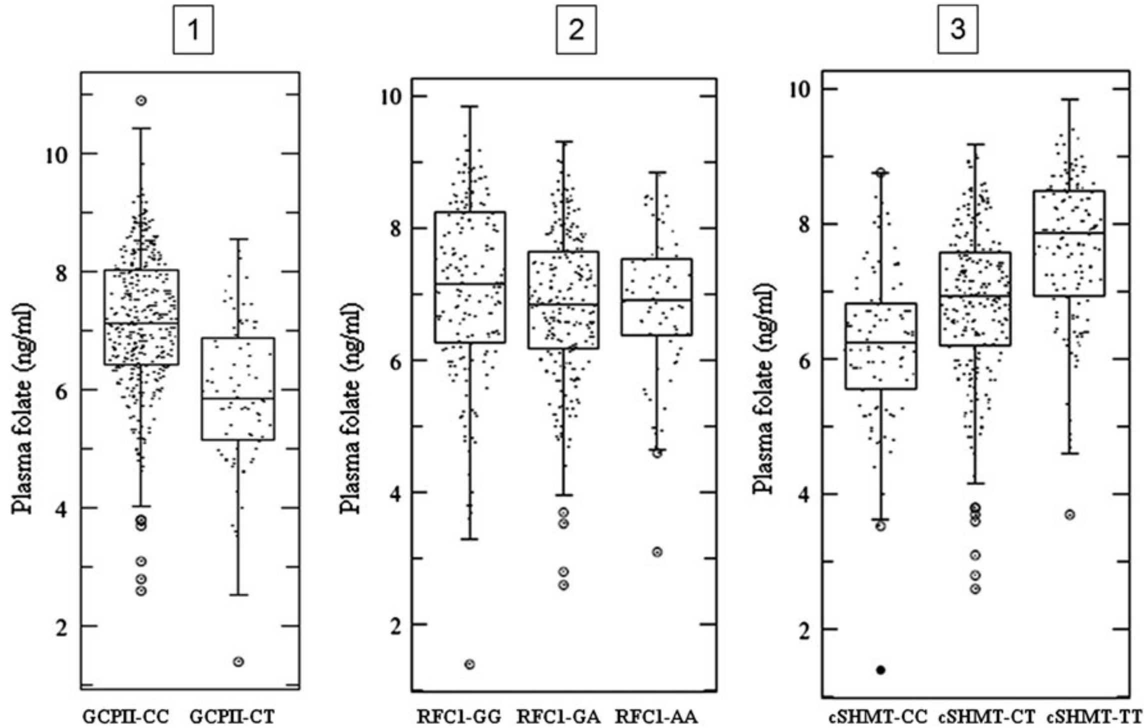


Fig. 1 Impact of GCPII C1561T, RFC1 G80A and cSHMT C1420T polymorphisms on plasma folate. This figure demonstrates the impact of three polymorphisms in crucial genes regulating intestinal absorption of folate, transport and maintenance of one-carbon homeostasis. *X*-axis: A, B and C represent wild, heterozygous and homozygous mutant genotypes respectively; *Y*-axis: Plasma folate concentration expressed in ng/ml

which could be due to aging or the higher levels of circulating estradiol [29] or prolonged exposure of mammary glands to the estrogens. Women with BMI in the optimal range were found to have reduced risk for breast cancer, consistent with established literature.

This is the first study, depicting RFC1 G80A polymorphism as a risk factor for the breast cancer. RFC1 transports folate into the cells and hence, is crucial in maintaining the intracellular concentrations of folate [30]. RFC1 G80A polymorphism impairs folate transport across the RBC membrane [31]. Under conditions of severe folate deprivation, RFC1 is down-regulated as an adaptive response in the MCF-7 breast cancer cell lines [32]. Since our subjects had low dietary folate intake, RFC1 G80A polymorphism coupled with low expression of RFC would induce severe RBC folate deficiency and hence might have increased the risk for breast cancer.

The risk associated with MTHFR C677T could be due to the thermolabile MTHFR enzyme that has enhanced propensity to lose the active dimer form with subsequent loss in the FAD-binding capacity and decreased specific activity [33]. The association studies related to breast cancer and MTHFR C677T were contradictory. Previous studies demonstrated an inverse association between the dietary folate intake [34, 35] or serum folate levels [36] and the breast cancer risk in subjects with MTHFR C677T polymorphism. Campbell et al. have reported the risk of early onset of the breast cancer in women with MTHFR C677T polymorphism [37]. Lee et al. have observed a non-significant association between MTHFR TT genotype and the breast cancer risk [7]. These studies indicate that nutritional status acts as an effect modifier for the MTHFR-associated breast cancer risk. Our study showed no such interaction between MTHFR and folate. TYMS 2R allele, which reduces TYMS expression 2–4 times [38], showed no independent effect on the breast cancer, which is in agreement with the other published studies [8].

The current study is in agreement with Cheng et al. in reporting the protective role of cSHMT C1420T polymorphism in breast cancer. Subjects with 1420CC genotype had low plasma folate as compared to the subjects with CT and TT genotype. Similar observation has been reported by Heil et al. [39]. cSHMT is B_6-dependent enzyme, which catalyzes the reversible conversion of serine and tetrahydrofolate to glycine and methylenetetrahydrofolate and irreversible conversion of methylenetetrahydrofolate to 5-formyltetrahydrofolate (futile folate cycle) [40]. Formation of 5-formyl THF may be critical in maintaining the one-carbon homeostasis, particularly during the rapidly proliferative stages of development. Thus, the inhibition of cSHMT may be a mechanism to prevent an unnecessary accumulation of 5-methylTHF at the expense of depriving one-carbon units for thymidylate and purine synthesis.

Although, TYMS 2R is not an independent risk allele, it was found to play an additive role in combination with high risk alleles at RFC1 and MTHFR loci in inflating the breast cancer risk, which could be due to the cumulative effect of uracil misincorporation into DNA and defective remethylation, which in turn correlated with two hallmarks of cancer i.e. genomic instability and aberrant methylation profile (global hypomethylation and focal hypermethylation). The interactions between these three variants are biologically plausible because RBC folate is the main form of functional folate, which is needed for 5,10-methylenetetrahydrofolate synthesis. MTHFR and TYMS compete for this substrate for the conversion to 5-methyltetrahydrofolate and for the synthesis of thymidylate, respectively.

The risk associated with low dietary folate intake might be due to the decrease in the reaction velocities of MTHFR, SHMT and TS enzymes. This is supported by the reported reduction in methylation rate, decrease in methionine and SAM levels, and increase in plasma homocysteine as observed in the mathematical model of Reeds et al. [41].

The overall data suggested the role of dietary as well as genetic factors of one-carbon metabolism in influencing breast cancer risk probably by affecting the synthesis of S-adenosylmethionine (SAM). Global hypomethylation of DNA [42] or decreased methylation of catechol estrogens to methoxy estrogens [43] might be the consequences of reduced availability of SAM, which can trigger breast carcinogenesis.

The major hallmark of this study was the application of MDR analysis apart from the conventional statistical tests to explore the best combination of genotypes from a given set of data, which facilitated the better understanding of complex biological interactions. By using matched case–control pairs, potential confounders such as age, ethnicity, geographical location and menopause status were controlled at the time of enrollment of the subjects. Other confounders such as body mass index, age at menarche, age at first full term pregnancy, parity, and family history of breast cancer were controlled using the logistic regression models. All the significant genetic associations had $\geq$80% power with type I α error.

The limitations of this study were: (*i*) The assessment of dietary intake of B vitamins after the breast cancer diagnosis sensitive to recall bias. (*ii*), The assessment of risk for heterozygous and homozygous mutants separately could not be done as limited number of subjects of the study were homozygous mutants for certain SNPs. (*iii*) As we recruited the subjects from only one hospital in Hyderabad, India, they all might not necessarily represent the general population, and thus any extrapolation of these results to the general population should be judged cautiously. (*iv*) Studies on different ethnic groups and populations warrant further corroboration.

To conclude, in this case–control study conducted on Indian women, we found low dietary folate intake, RFC G80A and MTHFR C677T as independent risk factors for the breast cancer, whereas cSHMT was found to be protective. TYMS 5′-UTR 2R allele, although exhibited no independent risk for the breast cancer, it was found to inflate the risk in presence of RFC1 80A and MTHFR 677T variants.

Acknowledgments This work was supported by the grant funded by Indian Council of Medical Research (ICMR), New Delhi (Ref No. 5/13/32/2007). We thank Miss Y. Rupasree, Mrs. S.V. Vijaya Lakshmi, Miss P. Shree Divyya and Mr. E. Chandra Sekhar for providing technical support.

References

1. Murthy NS, Chaudhry K, Nadayil D, Agarwal UK, Saxena S (2009) Changing trends in incidence of breast cancer: Indian scenario. Indian J Cancer 46:73–74
2. Rebbeck TR, Walker AH, Phelan CM, Godwin AK, Buetow KH, Garber JE, Narod SA, Weber BL (1997) Defining etiologic heterogeneity in breast cancer using genetic biomarkers. Prog Clin Biol Res 396:53–61 Review
3. Halapi E, Hakonarson H (2002) Advances in the development of genetic markers for the diagnosis of disease and drug response. Expert Rev Mol Diagn 2(5):411–421 Review
4. Ford D, Easton DF (1995) The genetics of breast and ovarian cancer. Br J Cancer 72:805–812
5. Melnyk S, Pogribna M, Miller BJ, Basnakian AG, Pogribny IP, James SJ (1999) Uracil misincorporation, DNA strand breaks, and gene amplification are associated with tumorigenic cell transformation in folate deficient/repleted Chinese hamster ovary cells. Cancer Lett 146(1):35–44
6. Christman JK, Sheikhnejad G, Dizik M, Abileah S, Wainfan E (1993) Reversibility of changes in nucleic acid methylation and gene expression induced in rat liver by severe dietary methyl deficiency. Carcinogenesis 14(4):551–557
7. Lee SA, Kang D, Nishio H, Lee MJ, Kim DH, Han W, Yoo KY, Ahn SH, Choe KJ, Hirvonen A, Noh DY (2004) Methylenetetrahydrofolate reductase polymorphism, diet, and breast cancer in Korean women. Exp Mol Med 36(2):116–121
8. Xu X, Gammon MD, Zhang H, Wetmur JG, Rao M, Teitelbaum SL, Britton JA, Neugut AI, Santella RM, Chen J (2007) Polymorphisms of one-carbon-metabolizing genes and risk of breast cancer in a population-based study. Carcinogenesis 28(7):1504–1509
9. Cheng CW, Yu JC, Huang CS, Shieh JC, Fu YP, Wang HW, Wu PE, Shen CY (2008) Polymorphism of cytosolic serine hydroxymethyltransferase, estrogen and breast cancer risk among Chinese women in Taiwan. Breast Cancer Res Treat 111(1):145–155
10. Akisik E, Dalay N (2007) Functional polymorphism of thymidylate synthase, but not of the COMT and IL-1B genes, is associated with breast cancer. J Clin Lab Anal 21(2):97–102
11. Chen J, Gammon MD, Chan W, Palomeque C, Wetmur JG, Kabat GC, Teitelbaum SL, Britton JA, Terry MB, Neugut AI, Santella RM (2005) One-carbon metabolism, MTHFR polymorphisms, and risk of breast cancer. Cancer Res 65(4):1606–1614
12. Grieu F, Powell B, Beilby J, Iacopetta B (2004) Methylenetetrahydrofolate reductase and thymidylate synthase polymorphisms are not associated with breast cancer risk or phenotype. Anticancer Res 24(5B):3215–3219

13. Lissowska J, Gaudet MM, Brinton LA, Chanock SJ, Peplonska B, Welch R, Zatonski W, Szeszenia-Dabrowska N, Park S, Sherman M, Garcia-Closas M (2007) Genetic polymorphisms in the one-carbon metabolism pathway and breast cancer risk: a population-based case-control study and meta-analyses. Int J Cancer 120(12):2696–2703
14. Pepe C, Guidugli L, Sensi E, Aretini P, D'Andrea E, Montagna M, Manoukian S, Ottini L, Radice P, Viel A, Bevilacqua G, Caligo MA (2007) Methyl group metabolism gene polymorphisms as modifier of breast cancer risk in Italian BRCA1/2 carriers. Breast Cancer Res Treat 103(1):29–36
15. Lin WY, Chou YC, Wu MH, Huang HB, Jeng YL, Wu CC, Yu CP, Yu JC, You SL, Chu TY, Chen CJ, Sun CA (2004) The MTHFR C677T polymorphism, estrogen exposure and breast cancer risk: a nested case-control study in Taiwan. Anticancer Res 24(6):3863–3868
16. Shrubsole MJ, Gao YT, Cai Q, Shu XO, Dai Q, Jin F, Zheng W (2006) MTR and MTRR polymorphisms, dietary intake, and breast cancer risk. Cancer Epidemiol Biomarkers Prev 15(3):586–588
17. Justenhoven C, Hamann U, Pierl CB, Rabstein S, Pesch B, Harth V, Baisch C, Vollmert C, Illig T, Brüning T, Ko Y, Brauch H (2005) One-carbon metabolism and breast cancer risk: no association of MTHFR, MTR, and TYMS polymorphisms in the GENICA study from Germany. Cancer Epidemiol Biomarkers Prev 14(12):3015–3018
18. Matsuo K, Hamajima N, Hirai T, Kato T, Inoue M, Takezaki T, Tajima K (2002) Methionine synthase reductase gene A66G polymorphism is associated with risk of colorectal cancer. Asian Pac J Cancer Prev 3(4):353–359
19. Gemmati D, Ongaro A, Scapoli GL, Della Porta M, Tognazzo S, Serino ML, Di Bona E, Rodeghiero F, Gilli G, Reverberi R, Caruso A, Pasello M, Pellati A, De Mattei M (2004) Common gene polymorphisms in the metabolic folate and methylation pathway and the risk of acute lymphoblastic leukemia and non-Hodgkin's lymphoma in adults. Cancer Epidemiol Biomarkers Prev 13(5):787–794
20. Shi Q, Zhang Z, Li G, Pillow PC, Hernandez LM, Spitz MR, Wei Q (2005) Polymorphisms of methionine synthase and methionine synthase reductase and risk of lung cancer: a case-control analysis. Pharmacogenet Genomics 15(8):547–555
21. Zhang Z, Shi Q, Liu Z, Sturgis EM, Spitz MR, Wei Q (2005) Polymorphisms of methionine synthase and methionine synthase reductase and risk of squamous cell carcinoma of the head and neck: a case-control analysis. Cancer Epidemiol Biomarkers Prev 14(5):1188–1193
22. Carr DF, Whiteley G, Alfirevic A, Pirmohamed M, FolATED study team (2009) Investigation of inter-individual variability of the one-carbon folate pathway: a bioinformatic and genetic review. Pharmacogenomics J 9(5):291–305 Review
23. Kumar K, Kaiser J (2006) Methylene tetrahydrofolate reductase (MTHFR) C677T and A1298C polymorphisms and breast cancer in South Indian population. Int J Cancer Res 2(2):143–151
24. Gopalan C, Rama Sastri BV, Balasubramanian SC (2007) Nutritive value of Indian foods. National Institute of Nutrition, Indian Council of Medical Research, India
25. Krebs J (2002) McCance and Widdowson's the composition of foods: summary edition, 6th summary ed. The Royal Society of Chemistry/Food Standards Agency, Cambridge/London
26. U.S. Department of Agriculture, Agricultural Research Service (2006) USDA National Nutrient Database for Standard Reference, Release 18. Nutrient Data Laboratory Home Page. http://www.ars.usda.gov/ba/bhnrc/ndl
27. Salazar LA, Hirata MH, Cavalli SA, Machado MO, Hirata RD (1998) Optimized procedure for DNA isolation from fresh and cryopreserved clotted human blood useful in clinical molecular testing. Clin Chem 44:1748–1750
28. Pike MC, Spicer DV, Dahmoush L, Press MF (1993) Estrogens, progestogens, normal breast cell proliferation, and breast cancer risk. Epidemiol Rev 15(1):17–35
29. Ursin G, London S, Stanczyk FZ, Gentzschein E, Paganini-Hill A, Ross RK, Pike MC (1999) Urinary 2-hydroxyestrone/16alpha-hydroxyestrone ratio and risk of breast cancer in postmenopausal women. J Natl Cancer Inst 91(12):1067–1072
30. Jansen G, Mauritz R, Drori S, Sprecher H, Kathmann I, Bunni M, Priest DG, Noordhuis P, Schornagel JH, Pinedo HM, Peters GJ, Assaraf YG (1998) A structurally altered human reduced folate carrier with increased folic acid transport mediates a novel mechanism of antifolate resistance. J Biol Chem 273(46): 30189–30198
31. Chango A, Emery-Fillon N, de Courcy GP, Lambert D, Pfister M, Rosenblatt DS, Nicolas JP (2000) A polymorphism (80 G $\rightarrow$ A) in the reduced folate carrier gene and its associations with folate status and hyperhomocysteinemia. Mol Genet Metab 70:310–315
32. Ifergan I, Jansen G, Assaraf YG (2008) The reduced folate carrier (RFC) is cytotoxic to cells under conditions of severe folate deprivation. RFC as a double edged sword in folate homeostasis. J Biol Chem 283(30):20687–20695
33. Yamada K, Chen Z, Rozen R, Matthews RG (2001) Effects of common polymorphisms on the properties of recombinant human methylenetetrahydrofolate reductase. Proc Natl Acad Sci USA 98(26):14853–14858
34. Sharp L, Little J, Schofield AC, Pavlidou E, Cotton SC, Miedzybrodzka Z, Baird JO, Haites NE, Heys SD, Grubb DA (2002) Folate and breast cancer: the role of polymorphisms in methylenetetrahydrofolate reductase (MTHFR). Cancer Lett 181(1): 65–71
35. Shrubsole MJ, Gao YT, Cai Q, Shu XO, Dai Q, Hébert JR, Jin F, Zheng W (2004) MTHFR polymorphisms, dietary folate intake, and breast cancer risk: results from the Shanghai Breast Cancer Study. Cancer Epidemiol Biomarkers Prev 13(2):190–196
36. Beilby J, Ingram D, Hähnel R, Rossi E (2004) Reduced breast cancer risk with increasing serum folate in a case-control study of the C677T genotype of the methylenetetrahydrofolate reductase gene. Eur J Cancer 40(8):1250–1254
37. Campbell IG, Baxter SW, Eccles DM, Choong DY (2002) Methylenetetrahydrofolate reductase polymorphism and susceptibility to breast cancer. Breast Cancer Res 4(6):R14
38. Horie N, Aiba H, Oguro K, Hojo H, Takeishi K (1995) Functional analysis and DNA polymorphism of the tandemly repeated sequences in the 5′-terminal regulatory region of the human gene for thymidylate synthase. Cell Structr Funct 20(3):191–197
39. Heil SG, Van der Put NM, Wass ET, den Heijer M, Trijbels FJ, Blom HJ (2001) Is mutated serine hydroxymethyltransferase (SHMT) involved in the etiology of neural tube defects? Mol Genet Metab 73(2):164–172
40. Fu TF, Hunt S, Schirch V, Safo MK, Chen BH (2005) Properties of human and rabbit cytosolic serine hydroxymethyltransferase are changed by single nucleotide polymorphic mutations. Arch Biochem Biophys 442(1):92–101
41. Reed MC, Nijhout HF, Neuhouser ML, Gregory JF III, Shane B, James SJ, Boynton A, Ulrich CM (2006) A mathematical model gives insights into nutritional and genetic aspects of folate-mediated one-carbon metabolism. J Nutr 136(10):2653–2661
42. Dizik M, Christman JK, Wainfan E (1991) Alterations in expression and methylation of specific genes in livers of rats fed a cancer promoting methyl-deficient diet. Carcinogenesis 12(7):1307–1312
43. Butterworth M, Lau SS, Monks TJ (1996) 17 beta-Estradiol metabolism by hamster hepatic microsomes. Implications for the catechol-*O*-methyl transferase-mediated detoxication of catechol estrogens. Drug Metab Dispos 24(5):588–594